Notes: Khwaja & Mian (AER, 2008)

Tracing the Impact of Bank Liquidity Shocks: Evidence from an Emerging Market

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1 Big picture

- **Question.** How do liquidity shocks to banks get transmitted to firms through the bank lending channel?
- **Empirical setting.** The paper studies Pakistan around the May 1998 nuclear tests. After the tests, the government froze foreign-currency deposit accounts. Banks that relied more on dollar deposits before the shock suffered larger deposit outflows afterward.
- **Main challenge.** A decline in bank lending could reflect lower loan demand from firms, not lower credit supply from banks.
- **Key identification idea.** Use firms that borrow from multiple banks and compare lending by different banks to the *same firm*. Firm fixed effects absorb the firm's credit demand shock.
- **Bank lending channel result.** A 1 percent decline in bank liquidity reduces lending to an existing borrower by about 0.6 percent in the preferred firm-fixed-effects specification.
- **Extensive-margin result.** Liquidity-constrained banks are more likely to lose existing borrowers and less likely to originate new loans.
- **Price result.** Interest rates do not rise significantly for liquidity-constrained banks, so the main adjustment is quantity rationing rather than higher prices.
- **Firm borrowing channel result.** Large firms can hedge bank-specific liquidity shocks by borrowing from other banks. Small firms cannot.
- **Financial distress result.** Small firms facing bank liquidity shocks experience higher default rates, consistent with real financial consequences.
- **Economic takeaway.** Bank liquidity shocks matter most when firms cannot substitute across lenders. The bank lending channel is therefore not just about banks; it depends on the borrower side of the credit market.

2 Data and measurement

2.1 Shock: the freeze of foreign-currency accounts

- Pakistan had allowed foreign-currency deposit accounts in the early 1990s.
- Dollar deposits grew rapidly before the 1998 nuclear tests.
- After the nuclear tests, the government froze foreign-currency accounts, triggering a sharp fall in dollar deposits.
- Banks differed in their pre-shock exposure to dollar deposits.
- This cross-bank heterogeneity generates the liquidity shock used in the paper.

Interpretation: The shock is useful because it is bank-specific: banks with more dollar deposits before the tests lose more liquidity after the freeze.

2.2 Loan data

- The main data come from the Credit Information Bureau of the State Bank of Pakistan.
- The data cover the universe of corporate loans outstanding in Pakistan between July 1996 and March 2000.
- The analysis focuses on 42 commercial banks that could take demandable deposits.
- A loan is defined as a bank–firm pair, with multiple loans from the same bank to the same firm aggregated.

- The intensive-margin sample contains 22,176 performing bank–firm loans to 18,647 firms.

Interpretation: The data are unusually powerful because the authors observe lending relationships at the bank–firm level, not only aggregate bank balance sheets.

2.3 Pre- and post-shock windows

The paper collapses quarterly data into one pre-shock and one post-shock observation:

$$\text{Pre period} : 1996 : III \text{ to } 1998 : I, \qquad \text{Post period} : 1998 : III \text{ to } 2000 : I.$$

Interpretation: Collapsing to two periods simplifies the event-study logic: measure how each loan changes from before to after the nuclear-test liquidity shock.

2.4 Main bank liquidity variable

For bank i, the bank liquidity shock is measured as

$$\Delta D_i = \Delta \log(\text{bank deposits}_i).$$

- A negative value means the bank loses deposits.
- The main source of variation is the bank's pre-shock share of dollar deposits.
- Figure 2 and Table 2 show that banks with higher dollar-deposit exposure experienced larger post-shock deposit declines.

Interpretation: A positive coefficient on ΔD_i in a lending regression means that a decline in bank deposits reduces lending.

2.5 Loan-level outcome variables

The main intensive-margin outcome is

$$\Delta L_{ij} = \Delta \log(\text{loan amount from bank } i \text{ to firm } j).$$

The extensive-margin outcomes are:

- **Exit**: an existing bank–firm relationship is not renewed.
- **Entry**: a new loan relationship is originated after the shock.

Interpretation: The intensive margin asks how much existing lending falls. The extensive margin asks whether firms lose bank relationships or fail to form new ones.

2.6 Firm-level borrowing and default outcomes

At the firm level, the authors aggregate borrowing across all 145 lending institutions:

$$\Delta \log(\text{aggregate firm borrowing}_j).$$

They also study firm financial distress using:

$$\Delta(\text{firm default rate}_j).$$

Interpretation: The bank lending channel matters for aggregate firm outcomes only if firms cannot replace lost credit from affected banks.

2.7 Firm heterogeneity

The paper emphasizes borrower size:

- **Small firms**: bottom 70 percent of the borrowing-size distribution.
- **Large firms**: top 30 percent of the borrowing-size distribution.

Other firm attributes include:

- political connection;
- membership in a business conglomerate;
- multiple banking relationships;
- city and industry controls.

Interpretation: Size is the key dimension of credit-market access. Large firms can substitute to other lenders; small firms are more locked into their initial banks.

3 Models and empirical specifications

3.1 Simple supply-demand model of bank lending

Banks can raise deposits costlessly up to their deposit base D_{it}. Beyond that, they must raise additional financing B_{it} with marginal cost

$$\alpha_B B_{it}, \qquad \alpha_B > 0.$$

Credit demand from firm j for bank i's loan is decreasing in loan size:

$$\bar{r}_j - \alpha_L L_{ijt}.$$

Interpretation: Costly external finance is what makes deposit shocks matter. If $\alpha_B = 0$, banks are in a Modigliani–Miller world and deposit shocks do not affect loan supply.

3.2 Bank and firm shocks

After the first period, bank i receives an economy-wide and bank-specific deposit shock:

$$D_{i,t+1} = D_{it} + \bar{d} + d_i.$$

Firm j receives an economy-wide and firm-specific productivity or credit-demand shock:

$$\bar{h} + h_j.$$

Interpretation: The empirical problem is that d_i and h_j may be correlated. If banks with deposit losses also lend to firms with falling demand, a simple lending regression confounds supply and demand.

3.3 Reduced-form change in lending

Solving the model gives

$$\Delta L_{ij} = \frac{\alpha_B}{\alpha_L + \alpha_B}(\bar{d} + d_i) + \frac{1}{\alpha_L + \alpha_B}(\bar{h} + h_j). \tag{1}$$

This can be rewritten as

$$\Delta L_{ij} = \beta_0 + \beta_1 \Delta D_i + h_j + \varepsilon_{ij}, \qquad \beta_1 = \frac{\alpha_B}{\alpha_L + \alpha_B}. \tag{2}$$

Interpretation: β_1 is the bank lending channel: how much a bank-specific liquidity shock changes lending to a borrower.

3.4 Naive OLS lending-channel regression

The OLS specification is

$$\Delta L_{ij} = \beta_0 + \beta_1 \Delta D_i + h_j + \varepsilon_{ij}. \tag{3}$$

If $\mathrm{corr}(\Delta D_i, h_j) \neq 0$, then

$$\widehat{\beta}_1^{OLS} = \beta_1 + \frac{\mathrm{cov}(\Delta D_i, h_j)}{\mathrm{var}(d_i)}.$$

Interpretation: OLS is biased because loan-level changes may combine bank credit supply and borrower credit demand.

3.5 Firm fixed effects

The preferred specification for the bank lending channel is

$$\Delta L_{ij} = \gamma_j + \beta_1 \Delta D_i + \varepsilon_{ij}, \tag{4}$$

where γ_j is a firm fixed effect.

Interpretation: This compares two banks lending to the same firm. Any firm-specific demand shock is absorbed by γ_j, so the remaining difference across the firm's banks identifies the bank supply effect.

3.6 Extensive-margin specification

For exits and entries, the paper estimates versions of:

$$Y_{ij} = \gamma_j + \beta_1 \Delta D_i + X_i' \theta + \varepsilon_{ij}, \tag{5}$$

where Y_{ij} is an exit or entry indicator and X_i contains bank controls.

Interpretation: A negative coefficient in the exit regression means that lower bank liquidity raises the probability that the relationship ends. A positive coefficient in the entry regression means lower liquidity reduces new relationship formation.

3.7 Firm borrowing channel

The firm-level aggregate equation is

$$\Delta Y_j = \beta_0^F + \beta_1^F \overline{\Delta D}_j + h_j, \tag{6}$$

where $\overline{\Delta D}_j$ is the loan-size-weighted average liquidity shock to firm j's pre-shock banks.

Interpretation: If firms can perfectly substitute to other lenders, then $\beta_1^F = 0$. If they cannot substitute, the firm borrowing channel is close to the bank lending channel.

3.8 Heterogeneity by firm size

The main heterogeneity specification is

$$\Delta Y_j = \beta_0 + \beta_1 \overline{\Delta D}_j + \beta_2 Small_j + \beta_3 (\overline{\Delta D}_j \times Small_j) + X_j' \theta + \varepsilon_j. \tag{7}$$

Interpretation: β_1 is the effect for large firms. $\beta_1 + \beta_3$ is the effect for small firms. The paper finds that large firms hedge but small firms cannot.

3.9 Financial distress and IV logic

For default, the paper estimates

$$\Delta Default_j = \beta_0 + \beta_1 \overline{\Delta D}_j + \beta_2 Small_j + \beta_3 (\overline{\Delta D}_j \times Small_j) + X_j' \theta + \varepsilon_j.$$

It also instruments the change in firm borrowing with the liquidity of the firm's banks.

Interpretation: If reduced borrowing causes distress, then bank liquidity shocks should raise firm default rates, especially for firms that cannot hedge.

4 Identification and empirical design

4.1 Why the shock is useful

- The nuclear tests and freeze of foreign-currency accounts create an aggregate event.
- Banks differ in pre-shock exposure to foreign-currency deposits.
- This produces cross-bank differences in deposit losses after the shock.
- The shock hits bank funding, not a specific firm relationship.

Interpretation: The event gives a plausibly exogenous shift in bank liquidity, but identification still requires separating bank supply from firm demand.

4.2 Why firm fixed effects are central

- Many firms borrow from multiple banks.
- For these firms, the authors compare lending by positive-liquidity and negative-liquidity banks to the same firm.
- The same firm has the same product demand, investment opportunities, and macro exposure across its bank relationships.
- Therefore, differences in lending changes across its banks are interpreted as bank credit-supply differences.

Interpretation: This is the cleanest part of the paper. It removes firm-level loan demand by construction.

4.3 Why OLS underestimates the effect here

- Normally, bank liquidity and borrower demand may be positively correlated.
- In this setting, the paper argues the cross-sectional correlation is negative.
- Dollar-reliant banks lost more deposits but initially lent to better-quality borrowers.
- Therefore, naive OLS tends to understate the true bank lending channel.

Interpretation: The comparison between OLS and firm-fixed-effects estimates supports this: the FE coefficient is larger than the corresponding OLS coefficient.

4.4 Demand and robustness concerns

The paper addresses several concerns:

- **Credit demand:** firm fixed effects absorb demand shocks at the firm level.
- **Loan type:** firm-by-loan-type fixed effects address loan-specific demand changes.
- **Bank quality:** bank controls include pre-shock ROA, capitalization, size, foreign/government status, and pre-shock default rates.
- **Price correction:** interest rates do not rise significantly for banks that lose liquidity.
- **Strategic borrower withdrawal:** affected banks were not becoming insolvent.

Interpretation: The empirical design is strongest for the loan-level bank lending channel; firm-level outcomes are interpreted more conservatively.

4.5 What the paper cannot fully prove

- Firm-level aggregate borrowing regressions cannot include firm fixed effects after aggregation.
- The paper cannot fully identify why large firms hedge better: size, bargaining power, relationships, collateral quality, and political or conglomerate connections may all matter.
- Interest-rate data are not available for every loan.
- The setting is Pakistan during an unusual event, so external validity depends on whether other banking systems have similar substitution frictions.

Interpretation: The main contribution is the identification of the lending channel and the distinction between bank-level and firm-level effects.

5 Tables and figures

5.1 Figures 1–2: the deposit shock and cross-bank exposure

Read:

- Figure 1 shows total dollar deposits rising steadily before the May 1998 shock and then falling rapidly afterward.
- Figure 2 shows that banks with higher pre-shock dollar-deposit shares experience lower deposit growth after the shock.
- The relationship in Figure 2 is bank-size weighted, making the pattern economically meaningful.

Economic message: The event creates the source of identifying variation: some banks lose much more liquidity because they were more exposed to foreign-currency deposits.

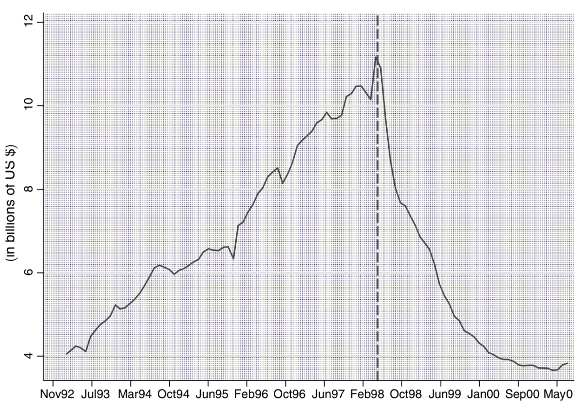


FIGURE 1. TOTAL DOLLAR DEPOSITS

(a) Total dollar deposits

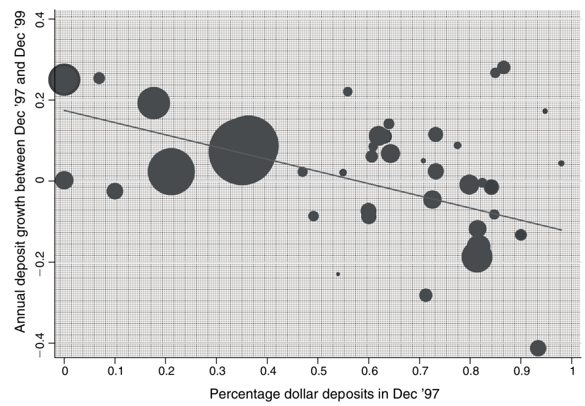


FIGURE 2. ANNUAL DEPOSIT GROWTH IN DEPOSITS AGAINST INITIAL DOLLAR DEPOSIT EXPOSURE (WEIGHTED)

(b) Deposit growth versus dollar-deposit exposure

5.2 Table 1: summary statistics

Read:

- The main loan-level sample contains 22,176 loans.
- Average pre-shock total lending per bank–firm pair is 16,479 in the paper's units.
- Average pre-shock interest rates are 15.9 percent.
- Large firms account for 94 percent of total lending but only 30 percent of firms.
- Multiple-relationship firms account for 66 percent of lending but only 10 percent of firms.
- Banks have an average dollar-deposit share of 60 percent and deposit-growth standard deviation of 30 percent.

Economic message: The distribution of credit is highly unequal: most firms are small, but most lending goes to large firms. This motivates the paper's borrower-size heterogeneity.

Table 1—Summary Statistics

Panel A: Loan-level variables (22,176 loans)				
Variable	Mean	S.D.		
Pre–nuclear test total lending (000)	16,479	60,768		
Change in log lending	−0.003	1.23		
Post–nuclear test default rate	6.8%	26.1%		
Pre–nuclear test interest rate (%)	15.9%	2.7%		
Loan type	*Fixed*	*Working capital*	*Letter of credit*	*Other*
Percent of total lending	32.5%	56.1%	4.2%	7.2%
Panel B: Borrower/firm attributes (18,647 firms)				
Politically connected	*No*	*Yes*		
Percent of total lending (of total firms)	54% (76%)	46% (24%)		
Size	*Small*	*Large*		
Percent of total lending (of total firms)	6.4% (70%)	94% (30%)		
Location (city size)	*Small*	*Medium*	*Large*	*Unclassified*
Percent of total lending (of total firms)	6% (14%)	13% (18%)	80% (62%)	6% (2%)
Multiple relationship	*Yes*	*No*		
Percent of total lending (of total firms)	66% (10%)	34% (90%)		
Business network size	*Non-conglomerate*	*Conglomerate*		
Percent of total lending (of total firms)	36% (85%)	64% (15%)		
Panel C: Bank-level variables (42 banks)				
Variable	*Mean*	*S.D.*		
Bank assets Dec '97	33886.3	63884.7		
Average ROA ('96 & '97)	0.013	0.027		
Capitalization rate ('96 & '97)	0.082	0.054		
Percentage of dollar deposits (Dec '97)	0.60	0.27		
Average default rate ('96 & '97)	0.086	0.13		
Growth in deposits (Dec '97 to Dec '99)	0.046	0.30		
Bank type	*Private*	*Foreign*	*Government*	
Percent of total lending	33.8%	36.8%	29.4%	

Notes: A "loan" is defined as a bank-firm pair, i.e., multiple loans of a firm from the same bank are aggregated up. The loan-level data comprise all performing loans given out by the 42 commercial banks at the time of nuclear testing that continued to be serviced. The pre and post data are averaged over June 1996 to March 1998, and June 1998 to March 2000, respectively. Note that since we include only performing pre-nuclear loans, the default rate just prior to nuclear

Table 1: Summary statistics

5.3 Table 2 and Figure 3: pre-shock exposure and aggregate lending trends

Read:

- Table 2 shows that a higher pre-shock dollar-deposit share predicts lower post-shock bank deposit growth.
- In the size-weighted regression, the coefficient is -0.30 with an R^2 of 0.40.
- Dollar-reliant banks have lower pre-shock default rates and higher pre-shock ROA, suggesting they were not weaker banks before the shock.
- Figure 3 shows that lending by positive- and negative-liquidity banks follows similar trends before the shock, then diverges after the shock.

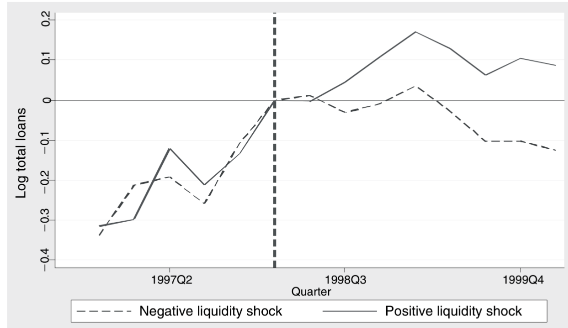
Economic message: The figure and table jointly support the supply-shock interpretation: banks exposed to frozen dollar deposits lost liquidity and reduced lending.

TABLE 2—BANK-LEVEL CORRELATIONS WITH PRETEST DOLLAR DEPOSIT EXPOSURE

Dependent variable	Average annual growth in bank deposits (Dec '99–Dec '97)		Average pre–nuclear test default rate		Average pre–nuclear test bank ROA	
	(1)	(2)	(3)	(4)	(5)	(6)
Percentage of deposits in dollars in Dec '97	−0.17	−0.30	−0.27	−0.31	0.044	0.061
	(0.08)	(0.06)	(0.06)	(0.06)	(0.014)	(0.016)
Constant	0.12	0.17	0.25	0.28	−0.013	−0.022
	(0.05)	(0.03)	(0.04)	(0.04)	(0.009)	(0.009)
Bank size weighted	No	Yes	No	Yes	No	Yes
Observations	42	42	42	42	42	42
R-squared	0.09	0.4	0.33	0.38	0.2	0.26

Notes: The regressions examine how dollar deposit reliant banks were affected by the liquidity shock—columns 1 and 2—and how they differed before—columns 3–6. The sample is the 42 commercial banks that were allowed to open dollar deposits and hence were directly affected by the "dollar freeze" as a result of the nuclear tests in May 1998. Average pre–nuclear test default rate is the loan-size weighted default rate of loans from a given bank from July 1996 to March 1998. The bank-level default rate is defined here as a fraction between 0 and 1. Average pre–nuclear test ROA is the average ROA of a bank over fiscal years 1996 and 1997 (years end in December). Robust standard errors in parentheses.

(a) Bank-level correlations with dollar-deposit exposure



(b) Aggregate bank lending channel

5.4 Table 3 and Figure 4: the bank lending channel

Read:

- Table 3 column 1 is the preferred firm-fixed-effects estimate: the coefficient on Δ log bank liquidity is 0.60.
- The interpretation is that a 1 percent fall in bank liquidity causes a 0.6 percent decline in lending to an existing borrower.
- Adding bank controls, lagged liquidity, loan-type interactions, and firm controls does not remove the effect.
- OLS estimates are smaller than firm-fixed-effects estimates, consistent with negative selection in this particular shock.
- The small-firm interaction is positive and large: small firms face a larger bank lending channel than large firms.
- Figure 4 repeats the logic within the same firm. Positive- and negative-liquidity banks behave similarly before the shock and diverge sharply afterward.

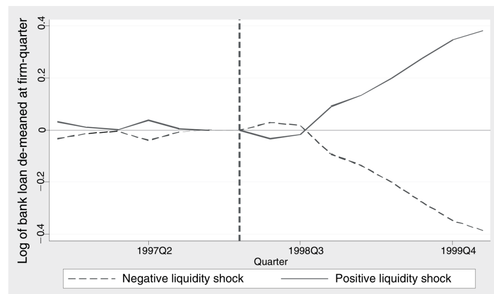
Economic message: The central result is a clean credit-supply effect. The same firm receives less credit from banks hit by larger liquidity shocks.

Table 3—The Bank Lending Channel—Intensive Margin

	Δ Log loan size						
Dependent variable	FE (1)	FE (2)	FE (3)	OLS (4)	OLS (5)	OLS (6)	OLS (7)
Δ Log bank liquidity	0.60	0.63	0.64	0.46	0.64	0.30	0.33
	(0.09)	(0.10)	(0.11)	(0.14)	(0.17)	(0.12)	(0.15)
Δ Log bank liquidity × small firms						0.57	0.40
						(0.26)	(0.21)
Small firms						0.18	0.24
						(0.06)	(0.03)
Lag Δ log bank liquidity		0.15					−0.13
		(0.10)					(0.14)
Preshock average bank ROA		0.99					−0.27
		(1.73)					(1.66)
Log bank size		0.02					−0.02
		(0.03)					(0.03)
Preshock bank capitalization		−1.16					0.09
		(0.97)					(1.13)
Preshock bank default rate		−0.869					−0.518
		(0.36)					(0.32)
Government bank dummy		0.13					−0.01
		(0.06)					(0.08)
Foreign bank dummy		0.01					−0.12
		(0.06)					(0.08)
Fixed effects	Firm	Firm	Firm × loan-type				Firm Controls
Constant	—	—	—	−0.06	−0.04	−0.14	—
				(0.04)	(0.04)	(0.03)	
Number of observations	5,382	5,382	5,382	5,382	22,176	22,176	22,176
R-squared	0.44	0.44	0.6	0.01	0.02	0.03	0.05

Notes: These regressions examine the bank lending channel for the set of firms borrowing at the time of the shock (the intensive margin) in more detail. All quarterly data for a given loan are collapsed to a single pre– and post–nuclear test period. The nuclear test occurred in the second quarter of 1998, so all observations from 1996:III to 1998:I for a given loan are time-averaged into one. Similarly, all observations from 1998:III to 2000:I are time-averaged into one. Data are restricted to: (a) banks that take retail (commercial) deposits (78 percent of all formal financing), and (b) loans that were not in default in the first quarter of 1998 (i.e., just before the nuclear tests). Columns 1–4 are run on the sample of firms that borrow from multiple banks (preshock) and include firm fixed effects (firm interacted with loan type for column 4). Columns 5–7 also include firms borrowing from single banks and run an OLS specification. Firm controls in column 7 include dummies for each of the 134 cities/towns the firm is located in, 21 industry dummies, whether the firm is politically connected, its membership in a business conglomerate, and whether it borrows from multiple banks.

(a) Intensive-margin lending channel



(b) Firm fixed effects graph

5.5 Tables 4–5: extensive margin and interest rates

Read:

- Table 4 shows that lower bank liquidity raises the likelihood of loan exit and lowers the likelihood of new loan entry.
- The paper summarizes the magnitude as follows: a one-standard-deviation liquidity shock leads to an 18 percent decline in lending, a 6.3 percentage point increase in exit, and a 3.6 percentage point decrease in new origination.
- Table 5 shows no significant increase in interest rates at liquidity-constrained banks.
- The absence of a rate increase suggests quantity rationing rather than an ordinary price adjustment.

Economic message: The lending channel works through both reduced loan amounts and relationship termination. Prices do not do the main adjustment.

Table 4—The Bank Lending Channel—Extensive Margin

Dependent variable	Exit?			Entry?		
	FE (1)	FE (2)	OLS (3)	FE (4)	FE (5)	OLS (6)
Δ Log bank liquidity	−0.21	−0.19	−0.16	0.12	0.15	0.087
	(0.05)	(0.05)	(0.059)	(0.05)	(0.04)	(0.049)
Small			0.084			0.2
			(0.019)			(0.015)
Small $\times$ Δ log bank liquidity			0.077			0.11
			(0.084)			(0.067)
Constant	—	—	—	—	—	—
Firm fixed effects	Yes	Yes		Yes	Yes	
Bank controls		Yes	Yes		Yes	Yes
Firm controls			Yes			Yes
Number of observations	6,517	6,517	26,730	8,516	8,516	35,921
R-squared	0.48	0.49	0.09	0.54	0.55	0.21

Notes: These regressions examine how the bank lending channel affected exit and entry of firms (from borrowing). Data are restricted to: (a) banks that take retail (commercial) deposits (78 percent of all formal financing), and (b) loans that were not in default in the first quarter of 1998 (i.e., just before the nuclear tests). Columns 1–3 look at exit by including all loans that were outstanding at the time of the nuclear tests. For a given loan, "exit" is classified as one if the loan is not renewed and the firm exits its banking relationship by the first postshock year. Columns 1–2 further limit the sample to only firms that were borrowing from multiple banks before the shock and include firm fixed effects. Columns 4–6 look at entry and include all loans given out after the nuclear tests quarter. For a given loan, "entry" is classified as one if the loan was made for the first time in the postshock year. Columns 4–5 further limit the sample to only firms that were borrowing from multiple banks after the shock and include firm fixed effects. All regressions include bank level controls: lagged change in bank liquidity, preshock bank ROA, log bank size, bank capitalization, fraction of portfolio in default, and dummies for foreign and government banks. The OLS regressions also include an extensive set of firm-level controls that include dummies for each of the 134 cities/towns the firm is located in, 21 industry dummies, whether the firm is politically connected, its membership in a business conglomerate, and whether it borrows from multiple banks. Standard errors in parentheses are clustered at the bank level (42 banks in total).

(a) Extensive margin: exit and entry

Table 5—Liquidity Impact on Interest Rates

Dependent variable	Δ Interest rate			
	(1) FE	(2) FE	(3) OLS	(4) OLS
Δ Log bank liquidity	0.28	0.33	1.53	−0.43
	(0.16)	(0.21)	(1.02)	(0.67)
Small firms				0.20
				(0.21)
Δ Log bank liquidity $\times$ small firms				0.64
				(0.78)
Fixed effects	Firm	Firm $\times$ loan type		
Bank controls				Yes
Firm controls				Yes
Constant	—	—	−1.59	
			(0.34)	
Number of observations	5,161	5,161	21,769	21,769
R-squared	0.43	0.57	0.02	0.13

Notes: These regressions examine the impact of the liquidity shock on interest rates. The interest rate data are not available for each loan, but rather at the bank branch level for different loan size classifications. Using a borrower's bank branch and loan size information, we can then create a "proxy" loan-level interest rate. All quarterly data for a given loan are then collapsed to a single pre– and post–nuclear test period. The nuclear test occurred in the second quarter of 1998, so all observations from 1996:III to 1998:I for a given loan are time-averaged into one. Similarly, all observations from 1998:III to 2000:I are time-averaged into one. Data are restricted to: (a) banks that take retail (commercial) deposits (78 percent of all formal financing), and (b) loans that were not in default in the first quarter of 1998 (i.e., just before the nuclear tests). Columns 1–2 further restrict the data to firms that were borrowing from multiple banks before the shock (in order to include firm fixed effects). Column 4 includes additional bank- and firm-level controls. The bank controls are the lagged change in bank liquidity, preshock bank ROA, log bank size, bank capitalization, fraction of portfolio in default, and dummies for foreign and government banks. Additional firm-level controls are dummies for each of the 134 cities/towns the firm is located in, and 21 industry dummies. Standard errors in parentheses are clustered at the bank level (42 banks in total).

(b) Interest-rate effects

5.6 Table 6 and Figure 5: the firm borrowing channel

Read:

- Table 6 shifts from bank–firm lending to total firm borrowing across all lenders.
- Column 1 shows an average firm borrowing channel of 0.65.
- Column 2 shows that large firms have a near-zero coefficient of 0.04, while the small-firm interaction is 0.80.
- Thus small firms face an aggregate borrowing decline of roughly 0.84 percent for a 1 percent decline in their initial banks' liquidity.
- Figure 5 plots the firm borrowing-channel coefficient by firm-size decile. The effect is large for smaller deciles and nearly absent for the largest deciles.

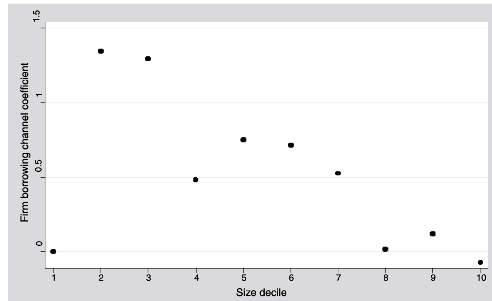
Economic message: Bank lending shocks matter for aggregate firm borrowing only when firms cannot substitute to other lenders. Large firms hedge; small firms do not.

Table 6—The Firm Borrowing Channel

Dependent variable	Δ Log aggregate loan size			
	OLS (1)	OLS (2)	OLS (3)	OLS (4)
Δ Log bank liquidity	0.65	0.04	0.00	0.29
	(0.04)	(0.09)	(0.09)	(0.11)
Small firms		0.18	0.19	0.28
		(0.02)	(0.02)	(0.03)
Δ Log bank liquidity × small firms		0.80	0.64	0.48
		(0.10)	(0.10)	(0.12)
Conglomerate firm?				0.09
				(0.03)
Δ Log bank liquidity × conglomerate firm				−0.28
				(0.14)
Political firm?				0.13
				(0.02)
Δ Log bank liquidity × political firm				−0.29
				(0.12)
Multiple relationship firms				0.18
				(0.03)
Δ Log bank liquidity × multiple relationship firms				−0.05
				(0.15)
Bank controls			Yes	Yes
Firm controls			Yes	Yes
Constant	0.04	−0.08	—	
	(0.01)	(0.02)		
Number of observations	18,647	18,647	18,647	18,647
R-squared	0.02	0.03	0.05	0.06

Notes: These regressions examine the impact of the liquidity shock on the total borrowing (across all lending institutions) of firms. All bank loans at a point in time (from any of the 145 lending institutions) for a given firm are summed to compute the aggregate firm-level loan size. The liquidity shock experienced by a firm is the (loan-size) weighted liquidity shock experienced by the banks it was borrowing from prior to the shock (lending institutions that do not hold deposits are assigned a liquidity shock of zero). All quarterly data for a given firm are then collapsed to a single pre– and post–nuclear test period. The nuclear test occurred in the second quarter of 1998, so all observations from 1996:III [illegible]

(a) Firm borrowing channel



(b) Firm borrowing channel by size decile

5.7 Tables 7–8: how large firms hedge and how small firms suffer

Read:

- Table 7 decomposes firm borrowing after the shock into borrowing from existing banks and new banks.
- For large firms, the coefficient on new banks is negative, meaning they borrow more from new banks when their old banks are hit.
- Small firms do not receive the same offsetting access to new banks.
- Table 8 shows that bank liquidity shocks increase default rates for affected firms, especially small firms.
- The IV specification implies a large effect of reduced firm loans on firm default rates.

Economic message: Large firms actively substitute across lenders. Small firms cannot, and the liquidity shock translates into financial distress.

TABLE 7—DECOMPOSING THE FIRM BORROWING CHANNEL

Dependent variable	Δ Log aggregate loan size aggregating loans post test using only:		
	Existing banks OLS (1)	New banks OLS (2)	Existing and new OLS (3)
Δ Log bank liquidity	0.15	−0.40	0.04
	(0.09)	(0.08)	(0.09)
Small firms	0.24	−0.23	0.18
	(0.02)	(0.02)	(0.02)
Δ Log bank liquidity × small firms	0.68	0.53	0.80
	(0.10)	(0.08)	(0.10)
Constant	−0.19	−2.55	−0.08
	(0.02)	(0.02)	(0.02)
Number of observations	18,647	18,647	18,647
R-squared	0.03	0.01	0.03

Notes: These regressions explore how firms compensate for their banks' liquidity shock. We split a firm's total borrowing postshock between banks it was borrowing from before the shock (column 1) and banks it started borrowing from after the shock (column 2). The liquidity shock experienced by a firm is the (loan-size) weighted liquidity shock experienced by the banks it was borrowing from prior to the shock. (Lending institutions that do not hold deposits are assigned a liquidity shock of zero.) All quarterly data for a given firm are collapsed to a single pre– and post–nuclear test period. The nuclear test occurred in the second quarter of 1998, so all observations from 1996:III to 1998:I for a given loan are time-averaged into one. Similarly, all observations from 1998:III to 2000:I are time-averaged into one. Data are restricted to loans that were not in default in the first quarter of 1998 (i.e., just before the nuclear tests). Standard errors in parantheses are clustered at the bank level, i.e., the largest lender for a firm. The bank controls are lagged change in bank liquidity, preshock bank ROA, log bank size, bank capitalization, fraction of portfolio in default, and dummies for foreign and government banks. Firm-level controls include dummies for each of the 134 cities/towns the firm is located in, 21 industry dummies, whether the firm is politically connected, its membership in a business conglomerate, and whether it borrows from multiple banks. Standard errors in parentheses are clustered at the bank level, i.e., the largest lender for a firm.

(a) Decomposing the firm borrowing channel

TABLE 8—FIRM BORROWING CHANNEL IMPACT ON FIRM FINANCIAL DISTRESS

Dependent variable	Δ Firm default rate				
	OLS (1)	IV (2)	OLS (3)	OLS (4)	OLS (5)
Δ Log bank liquidity	−13.71		2.01	−2.36	−4.84
	(7.44)		(3.46)	(3.07)	(3.80)
Δ Log firm loan		−45.45			
		(12.45)			
Small firms			3.61	1.09	0.91
			(1.18)	(0.90)	(0.91)
Δ Log bank liquidity × small firms			−18.50	−13.62	−11.50
			(4.57)	(3.99)	(3.83)
Conglomerate firm?					−3.41
					(0.56)
Δ Log bank liquidity × conglomerate firm					10.15
					(2.21)
Political firm?					−1.16
					(0.58)
Δ Log bank liquidity × political firm					−2.27
					(1.44)
Multiple relationship firms					−1.42
					(0.85)
Δ Log bank liquidity × multiple relationship firms					−0.11
					(2.77)
Bank controls				Yes	Yes
Firm controls				Yes	Yes
Constant	8.30	5.14	5.41	—	—
	(1.35)	(0.75)	(0.77)		
Number of observations	18,647	18,647	18,647	18,647	18,647
R-squared	0.01		0.02	0.05	0.05

Notes: These regressions examine the impact of the liquidity shock on the firm's average default rate. All bank loans at a point in time (from any of the 145 lending institutions) for a given firm are aggregated at the firm level to compute firm default rate, loan size, etc. The liquidity shock experienced by a firm is the (loan-size) weighted liquidity shock experienced by the banks it was borrowing from prior to the shock (lending institutions that do not hold deposits are assigned a liquidity shock of zero). All quarterly data for a given firm are then collapsed to a single pre– and post–nuclear test period. The nuclear test occurred in the second quarter of 1998, so all observations from 1996:III to 1998:I for a given loan are time-averaged into one. Similarly, all observations from 1998:III to 2000:I are time-averaged into one. Data

(b) Firm financial distress

6 Short summary

- The paper studies how bank liquidity shocks affect firm borrowing.
- The setting is Pakistan after the 1998 nuclear tests, when foreign-currency accounts were frozen.
- Banks with more pre-shock dollar deposits experienced larger deposit declines.
- The key identification strategy compares different banks lending to the same firm.
- Firm fixed effects remove firm-level demand shocks.
- The preferred bank lending-channel estimate is about 0.60: a 1 percent liquidity decline reduces bank lending by 0.6 percent.
- Liquidity-constrained banks are more likely to end existing relationships and less likely to start new relationships.
- Interest rates do not rise significantly, so banks ration credit mainly through quantities.
- Large firms can offset bank-specific liquidity shocks by borrowing from other banks.
- Small firms cannot hedge and suffer nearly the full aggregate borrowing decline.
- Large firms hedge partly by borrowing from new banks after the shock.
- Small firms exposed to bank liquidity shocks have higher default rates.
- The paper shows why the bank lending channel has heterogeneous real effects: credit-market substitution matters.

7 Conclusion

7.1 Core conclusion

Khwaja and Mian show that a bank liquidity shock reduces bank lending, but the real effect depends on the borrower. The same firm borrows less from banks hit by liquidity shocks, proving a supply-side bank lending channel. Yet large firms can substitute to other lenders, while small firms cannot.

7.2 Economic intuition

The intuition is that banks have costly external finance. When a bank loses deposits, it cannot costlessly replace that funding, so it cuts lending. If a firm can access other banks, the shock is only bank-specific. If the firm cannot access other banks, the bank-specific shock becomes a firm-level credit shock.

7.3 Takeaway

The paper's main methodological contribution is the use of firm fixed effects in bank–firm loan data to identify credit supply. Its main economic contribution is the distinction between the bank lending channel and the firm borrowing channel: bank shocks matter for the real economy when borrowers are unable to hedge across lenders.